

Deep Learning in Healthcare: Skin Cancer Detection

1. Introduction

Every year we are seeing significant increase in skin cancer cases globally, making it one of the most common types of cancer. If the skin cancer can be detected in a very early stage, chances of saving a patient increase. Nevertheless, following diagnosis methods like visual assessment by a doctor can be time-consuming and in most of the cases imprecise due to human error [1]. Deep learning can help in this scenario, the various advance models can precisely work on skin lesion images and accurately identify the cases of skin cancer like melanoma.

This report outlines skin cancer detection using deep learning models, discuss these techniques in detail and explore the advantages and disadvantages of models.

2. Current Deep Learning Techniques for Skin Cancer Detection

For image classification tasks, deep learning models – particularly Convolutional Neural Networks (CNNs) are preferred. These models are very effective in identifying skin cancer from dermoscopic images. Below is an overview of the best techniques used for skin cancer detection.

2.1 CNN-based Models

Convolutional Neural Networks (CNNs) are the backbone of most modern image classification tasks, including skin cancer detection. Several variations of CNNs have been developed to improve accuracy:

- **LeNet-5:** LeNet-5, created in 1998, was one of the earliest developed convolutional neural networks but was mainly for identifying handwritten numbers. Even though it was not made for detecting skin diseases, its architecture really helped in shaping medical imaging model later.
- **AlexNet:** AlexNet's 2012 victory in the ImageNet competition marked a major shift in computer vision. Its use of ReLU activation and dropout significantly improved performance, and these innovations later influenced how medical image analysis models were designed—especially in tasks like skin cancer classification, where overfitting is a concern.
- **VGGNet:** VGGNet, introduced in 2014, is famous for its structure that uses small 3x3 convolutional filters. It created a simple and strong system for classifying images, which help in examining skin lesions as it can identify small details.

2.2 Transfer Learning

Transfer learning is a useful approach where a pre-trained model which is already trained on large dataset is picked, and tweaked for a specific task like skin cancer detection which have a

limited dataset. For example, popular models like VGG16, VGG19, ResNet50, and InceptionV3, are already trained on ImageNet dataset, which is very large dataset, these models can be called and adjusted to work on limited dermatology images (such as skin lesion images).

Pre-trained models: Pre-trained models such as MobileNetV2, can be used in image classification tasks due to its accuracy and performance.

Fine tuning: Also in the fine tuning, to identify the unique patterns in skin lesion images, final layers of these models can be optimized so that a better accuracy can be obtained in diagnosing conditions like melanoma.

2.3 GAN-based Models

Generative Adversarial Networks (GANs) are useful in creating fake medical images that will help the deep learning models in training, this helps when we are short of labeled images. They have helped in producing a wide variety of skin lesion images, which helps the models to perform better and provide better accuracy. Two of the most popular type of GANs are CycleGAN and Pix2Pix.

2.4 Model Evaluation

In image classification, one the major issue is the model performance. For skin cancer detection, below are the evaluation metrics:

- **Accuracy:** This shows the percentage of correctly classified images.
- **Precision, Recall, and F1-Score:** These metrics are crucial in medical settings because getting a false positive or negative can lead to serious problems.
- **AUC-ROC Curve:** This is used to assess the balance between sensitivity and specificity.

2.5 Challenges and Limitations

While deep learning models have shown impressive results in skin cancer detection, still several challenges remain:

- **Data Quality and Size:** It is very difficult to get large labeled image data. Deep learning models require a large labeled dataset to detect the skin cancer correctly. The most commonly used datasets are ISIC 2018 dataset and HAM10000.
- **Class Imbalance:** An imbalance is created since malignant lesions are less common than the benign, this impacts the model's performance and accuracy.
- **Interpretability (Black box):** Doctors often find it hard to trust AI decisions because they don't know how these models make their choices. We need a better system that explains the reasoning behind the decisions, helping doctors understand how the AI came to its conclusions.

3. Industry Applications of Deep Learning

3.1 Healthcare

Deep learning is helping in analyzing patient's conditions, identifying diseases in early stages. It can be used for:

- **Detecting diseases from images:** Models can identify diseases through various images such MRIs, CT scans etc.
- **Early detection of diseases:** These models can help in identifying diseases at an early stage, which increases the chance of patient's life.
- **Personalized Medicine:** When the models are provided with patient's data, it can effectively provide treatment plans to each person.

3.2 Transportation

Deep learning has a positive impact on transportation industry:

- **Autonomous vehicles:** In self-driving cars, deep learning models analyze real-time information from camera and other sensors to perform various tasks like keeping the car in correct lane, avoiding crashes, detection pedestrians, traffic signals and stop signs. These models continuously train from the collected real-time data to improve its decision-making ability.
- **Managing traffic:** These models analyze patterns from past collected traffic data, GPS data etc. and can forecast accurately traffic jams. Based on this, it can provide better route option to drivers, making transportation much safer.

3.3 Security

Deep learning has transformed the security systems:

- **Intelligent surveillance:** Deep learning in precise facial recognition and make decision based on behavioral data, this could help in spotting an unauthorized person. Models can also find anomalies like unattended objects in high security areas, which eventually help in keeping security in place.
- **Cybersecurity:** Deep learning can identify fraud financial transactions, or a new malware in network traffic, keeping the security system safe from outside attack.

4. Future Advancements in Deep Learning for Skin Cancer Detection

4.1 Current challenges

While there is a significant progress in skin care detection, still there are some challenges:

- **Model generalization:** Deep learning models are mostly trained on curated datasets and usually underperform when a new diverse patient's datasets are given which includes uncommon dermatology images etc. from different sources.

- **Data Privacy:** Since patient data are highly confidential, this makes it very difficult to get all new data for further training to improve model's decision-making ability.
- **Lack of transparency:** Often it is tough for doctors to trust AI decisions, since they are unaware how these models take decisions. A better system needs to be implemented where model will explain the reasoning in a better way and help understand the doctors how they have reached to a decision.

4.2 Potential Solutions

- **Data augmentation & synthetic generation:** Using generative adversarial network, we can create synthetic data, which will help in reducing bias in the training data and help the model make better decisions when it is provided with new data.
- **Adaptive transfer learning framework:** When pre-trained models are provided with small dataset, it can still identify skin cancer condition accurately [2].
- **Interpretable AI for clinical trust:** Latest methods are helping doctor understand how a model has reached to some decisions. Grad-CAM ++ helps in creating heatmaps that will highlight cancer areas in the images. Along with this, ProtoPNet collects past records and make decisions in identifying new medical cases.

4.3 The Future

- Merging CNNs with RNNs can help recognize both spatial and time-related patterns in images. Basically, it's like giving the model a better memory and focus, so it can pick up on both the details in the image and any patterns over time.
- Deep learning models can work with devices like dermatoscopes for real time diagnosis.
- If the deep learning models are placed in mobile devices, models can detect skin cancer more easily.

5. Code Experiment: Skin cancer classification Using CNNs

We also experimented skin cancer classification using HAM10000 dataset.

5.1 Dataset Formation and preprocessing

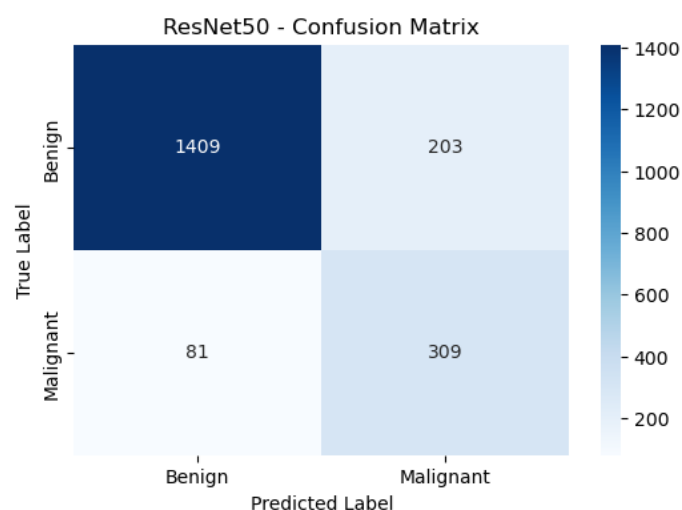
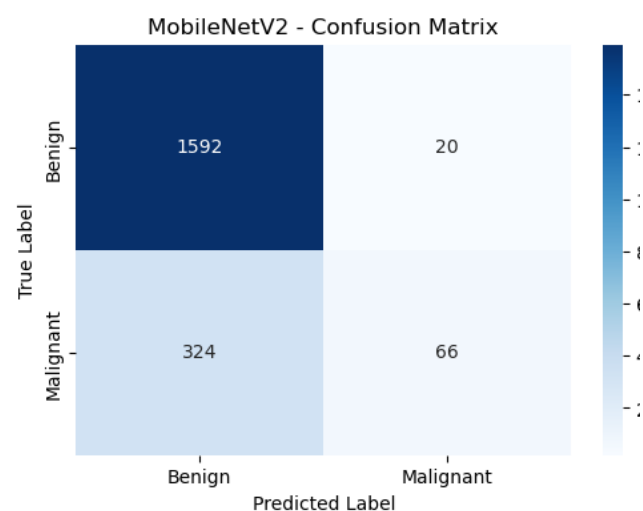
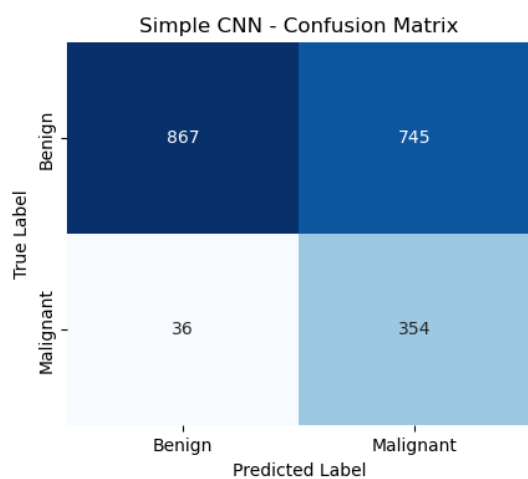
The dataset contains 10,000+ dermoscopic images. Images contains two categories: one is benign lesions and other is malignant lesions. Also in data preprocessing, data augmentation and imbalance were handled.

5.2 Model Architecture & Results

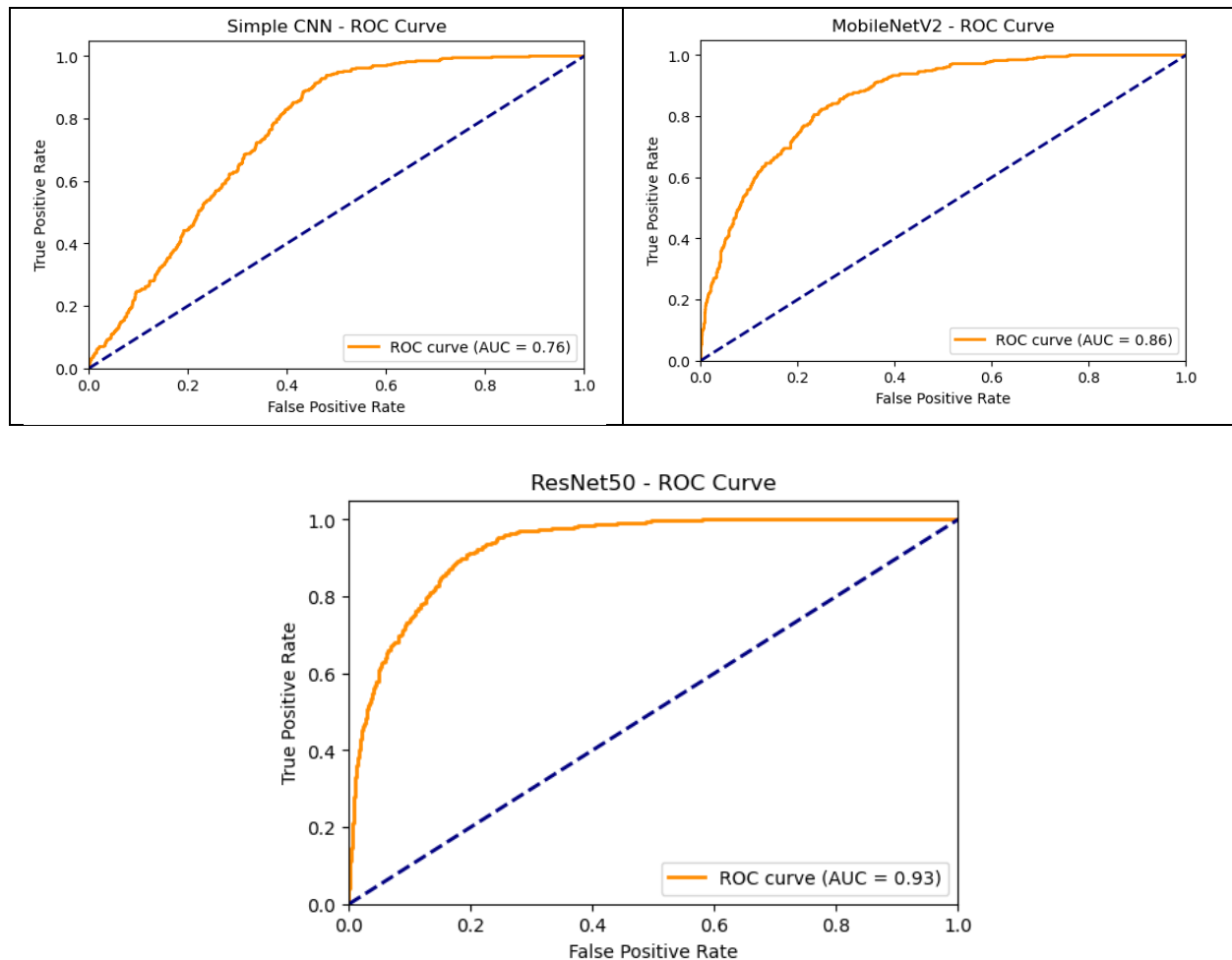
We used three models: Simple CNN, MobileNetV2 and ResNet50. They all utilized binary cross-entropy loss, the Adam optimizer, and early stopping.

Model	Accuracy	Benign Precision	Malignant Precision	Malignant Recall	F1 Score (Malignant)
Simple CNN	61%	0.96	0.32	0.92	0.48
MobileNetV2	83%	0.83	0.76	0.17	0.27
ResNet50	86%	0.94	0.61	0.78	0.69

As per the above results and confusion matrix analysis, ResNet50 performed the best, out of 390 malignant skin lesions, it identified 309 correctly. ResNet50 works well to identify the different benign and malignant cases. With this MobileNetV2 can only find 66 correct malignant cases. And the last simple CNN model performed the lowest of all, many benign cases were labeled as malignant.



Also, when comparing ROC curves with their AUC scores, again ResNet50 comes out to be the best, as it got the highest AUC. ResNet50 was able to distinguish both benign and malignant cases.



6. Conclusion

Using advanced deep learning models can help greatly in skin care detection and in healthcare general. Model's accuracy and efficiency can help in finding and analyzing the diseases in very early stage. With many advancements, deep learning models is still facing many challenges such as data availability, data transparency etc. Deep learning could become a valuable resource for doctors globally in detecting various diseases.

Reference:

- [1] Esteva, A., Kuprel, B., Novoa, R. A., Ko, J., Swetter, S. M., Blau, H. M., & Thrun, S. (2017). *Dermatologist-level classification of skin cancer with deep neural networks*. *Nature*, 542(7639), 115–118. <https://doi.org/10.1038/nature21056>
- [2] C. Menegola et al., "Knowledge Transfer for Melanoma Screening with Deep Learning," in 2017 IEEE 14th International Symposium on Biomedical Imaging (ISBI 2017), pp. 297–300.